01
COVER

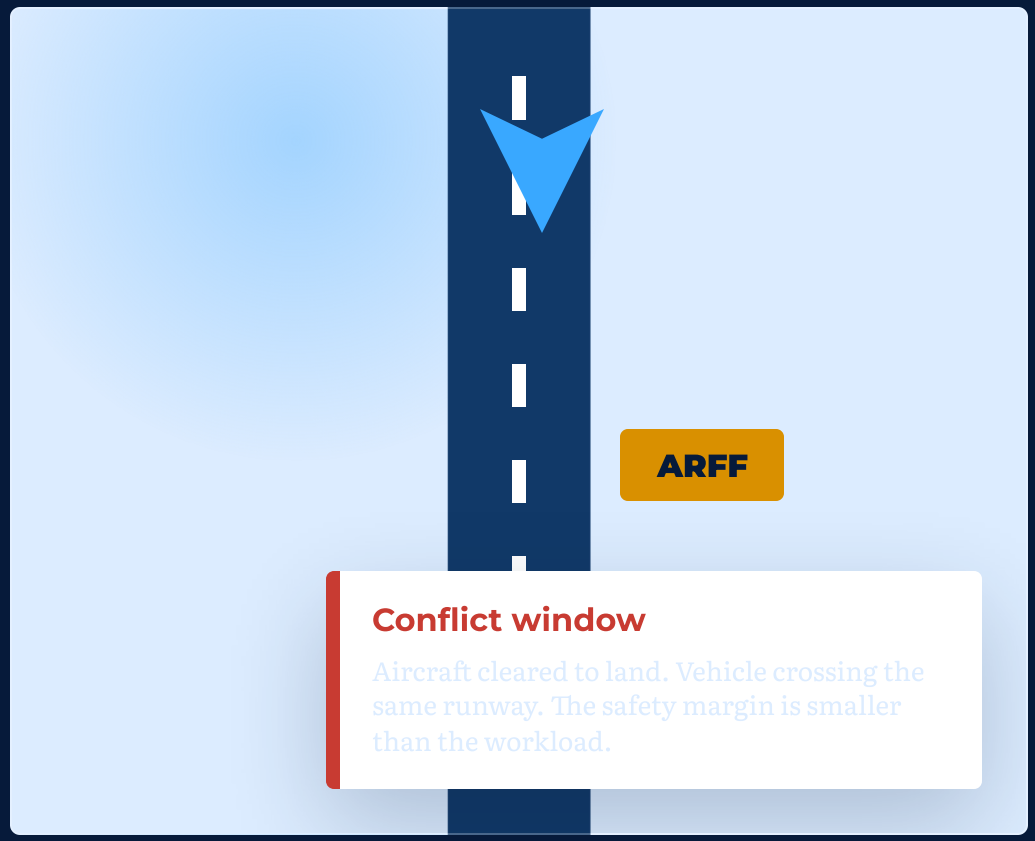
ADVANCED TECHNOLOGY AND INNOVATION

# ATC FlowLens

Explainable runway safety and ATC decision-support dashboard for tower, ground, training, and safety review.

**A clearer operational lens when runway decisions are measured in seconds.**

Team: Agentic Vibing | Simulation-first MVP | Human-in-the-loop




BUILD Indonesia
Aerospace & Defense
Agentic Vibing

02

PROBLEM

SLIDE 2 - PROBLEM

# Controllers must make fast decisions while monitoring many moving parts at once.

ATC workload rises when traffic, voice communication, weather, runway crossing, vehicle movement, and coordination happen at the same time.

### What must be monitored

Aircraft approaching, landing, departing, or taxiing; active runway status; airport vehicles; runway crossing requests; weather and visibility; radio calls; readbacks; flight strips; and unit coordination.

### Why workload becomes unsafe

Critical information may be distributed across displays, voice radio, human memory, and procedure assumptions. When time-to-conflict is short, the full picture must be understood immediately.

### The runway conflict gap

A landing or departing aircraft can conflict with a ground vehicle if clearance status, vehicle movement, and aircraft timing are not interpreted as one operational state.

**The opportunity is not to replace controllers, but to help them connect runway risk faster when workload is already high.**

Runway operations
are time-critical

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EVIDENCE

SLIDE 3 - EVIDENCE

# Runway conflicts and workload-related incidents have happened in real operations.

**345**

runway incursions across FAA Core 30 airports in FY2024.

**391**

average annual Core 30 incursions before the pandemic, FY2017-FY2019.

**12**

Core 30 airports exceeded pre-pandemic incursion levels in FY2024.

**DFW/BOS/ORD**

had the highest Core 30 incursion counts in FY2024.

| Case | What happened | Relevance to ATC FlowLens |
|---|---|---|
| **LATAM Peru 2213, 2022** | A320neo collided with a fire truck on runway during takeoff roll in Lima. | Shows aircraft-vehicle conflict during emergency vehicle activity and coordination gaps. |
| **Austin-Bergstrom, 2023** | Departure clearance occurred while another aircraft was on final approach in dense fog. | Shows how low visibility and runway status awareness reduce safety margin. |
| **LaGuardia, 2026 preliminary** | NTSB opened an investigation after a CRJ-900 collided with an ARFF vehicle on runway 4. | Shows why vehicle tracking, clearance timing, and workload matter. |

Real runway safety issue

Sources: FAA Air Traffic by the Numbers FY2024; NTSB DCA26MA161; BEA LATAM 2213 event page; AP/NTSB Austin coverage.

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GAP

SLIDE 4 - CURRENT SYSTEMS AND GAP

# ATC already has important technologies, but the challenge is connecting them into one clear operational picture.

### Existing ATC / ATM technologies

Radar, ADS-B, MLAT, surface surveillance, electronic flight strips, A-SMGCS, RIMCAS/RMCA, STCA/MTCD, CPDLC/Data Comm, AMAN/DMAN, TFDM, and SWIM.

### Coverage and fragmentation reality

Advanced surface tools are valuable but expensive and uneven. Voice communication remains common in tower and ground operations, so clearance state and runway timing still require interpretation.

### Existing data

Aircraft, vehicles, runway, clearance, weather, communication logs.

→

### FlowLens layer

Time-to-conflict, workload score, event explanation, replay timeline.

→

### Human decision

Controller remains in command. The tool supports awareness and training.

Current systems still fragment context

Sources: ICAO Global Runway Safety Action Plan; MITRE surface safety prototypes; SKYbrary RIMCAS; FAA AIM ASDE-X/ASSC.

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SOLUTION

SLIDE 5 - SOLUTION

# ATC FlowLens brings runway movement, clearance status, vehicle activity, communication, and workload risk into one dashboard.

The product does not automate ATC. It connects operational signals into a single timeline so teams can detect, explain, and replay runway risk.

### Runway occupancy

Clear, occupied, crossing, takeoff, or inbound landing state.

### Clearance state engine

Cleared to land, takeoff, hold short, cross, line up and wait, go around.

### Time-to-conflict

Compares aircraft runway entry or touchdown time with vehicle clear time.

### Communication timeline

Organizes tower, pilot, and vehicle messages into event history.

### Workload indicator

Scores traffic, radio calls, clearances, vehicles, weather, and alerts.

### Replay and explanation

Shows what happened, when it happened, and why an alert appeared.

One operational risk view

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HOW IT WORKS

SLIDE 6 - HOW IT WORKS

# The system converts simulated operational data into clear runway risk insight.

1

### Aircraft inbound

BAW123 is 1.8 NM final, speed 145 kt, cleared to land RWY 27.

2

### Vehicle crossing

ARFF-2 is cleared or starts to cross RWY 27 during active runway operation.

3

### Prediction

Touchdown estimated in 42 seconds; vehicle clear time estimated in 58 seconds.

4

### Conflict window

Vehicle clear time is later than aircraft touchdown time. Predicted overlap: 16 seconds.

5

### Explainable output

Alert severity, reason, relevant objects, workload state, and replay timeline.

### Example alert

HIGH RUNWAY CONFLICT RISK: BAW123 cleared to land RWY 27 while ARFF-2 is cleared to cross the same runway. Vehicle clear time exceeds aircraft touchdown time.

### Pitch value

The demo has visible inputs, processing, and outputs: aircraft/vehicle data, clearance state, time-to-conflict, explanation, workload, and replay.

Simulated data
to risk insight

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MVP

SLIDE 7 - MVP SCOPE AND DEMO PLAN

# The MVP is feasible for a hackathon because it uses simulation, rule-based logic, and a focused runway scenario.

### Hackathon MVP

One simulated airport layout, runway and taxiway, three to five aircraft, one to two vehicles, clearance logs, communication logs, weather or visibility status, conflict detection, workload scoring, and replay.

### Why this is feasible

It avoids live ATC data, regulatory integration, and certification dependency while still showing a realistic safety logic and operational dashboard.

### Why this is valuable

Training centers and airport safety teams need explainable scenarios, not only alarms. FlowLens becomes a reusable safety learning environment.

| Layer | What we build now | What it can become |
|---|---|---|
| **Data** | Scenario JSON and simulated positions. | ADS-B, airport operation data, weather, simulator logs. |
| **Logic** | Rule-based conflict detection and workload score. | Richer prediction, scenario generator, human factors model. |
| **Interface** | Runway map, alert reason, event replay. | Training lab, safety analytics, ANSP/aviation school collaboration. |

Focused demo buildable scope

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USERS

SLIDE 8 - TARGET USERS AND BUSINESS IMPACT

# The first adoption path is training, simulation, and safety awareness before real operational integration.

### Problem owners

Airport operators, Air Navigation Service Providers, ATC training centers, aviation academies, airport safety teams, and airport rescue and firefighting units.

### Adoption path

Training and simulation tool → safety briefing and scenario replay product → airport operation support prototype → research collaboration → future certified decision-support exploration.

## Training

Better runway conflict scenarios and workload awareness.

## Review

Faster explanation of complex event timelines.

## Readiness

Safer coordination between tower, ground, and emergency units.

## Expansion

Speech-to-text, readback mismatch, richer analytics.

**A simulation-first product is more fundable and buildable than a premature claim to replace certified ATC infrastructure.**

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IMPACT

SLIDE 9 - EXPECTED IMPACT AND BUILD INDONESIA ALIGNMENT

# ATC FlowLens aligns with BUILD Indonesia by applying advanced technology to a real aerospace safety challenge.

| Impact area | Expected value | BUILD alignment |
| --- | --- | --- |
| **Runway safety** | Recognize conflict risk earlier through aircraft, vehicle, clearance, and timing context. | Aerospace & Defense: airport operational readiness and aviation safety. |
| **Training and review** | Make conflict scenarios easier to explain and replay. | Advanced Technology and Innovation: digital support, simulated tracking, explainable alerting. |
| **Workload awareness** | Score traffic, radio calls, weather, vehicle movement, clearances, and active alerts. | Social Impact: safer operations protect passengers, crews, ground staff, emergency units. |
| **Operational flow** | Future versions may reduce avoidable go-around, taxi delay, holding, and fuel burn. | Sustainability: better runway flow can reduce waste in future operations. |

### Primary focus area

Advanced Technology and Innovation through prediction, workload scoring, and explainable runway visualization.

### Secondary relevance

Social Impact and Sustainability through safer runway operations and future reduction of avoidable fuel waste.

### Why it is finalist-worthy

Specific problem, real evidence, feasible MVP, careful safety positioning, and clear adoption path.

BUILD Indonesia alignment

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ROADMAP

SLIDE 10 - ROADMAP AND CLOSING

# ATC FlowLens starts as a feasible hackathon MVP and can grow into a broader aviation safety training platform.

### Phase 1: Hackathon MVP

Simulated runway map, aircraft and vehicle movement, clearance log, time-to-conflict alert, workload indicator, and replay timeline.

### Phase 2: Enhanced prototype

ATC-style speech-to-text, readback mismatch detection, scenario generator, multi-runway support, and richer workload model.

### Phase 3: Future vision

Airport operation datasets, tower-ground training simulator, safety analytics dashboard, and research pathway toward certified support.

**BUILD Indonesia can help validate the concept with mentors, aerospace expertise, and a stronger path from prototype to aviation safety product direction.**

### Final statement

ATC FlowLens is not trying to automate the controller. It gives controllers, trainers, and safety teams a clearer lens on runway flow, workload, and conflict risk when every second matters.